

CSC490H5 F

Capstone Design Course

Fall 2026 Syllabus

Course Meetings

CSC490H5 F

Section	Day & Time	Delivery Mode & Location
LEC0101 Machine Learning for Vision	Tuesday, 11:00 AM - 1:00 PM	In Person: DH 2020
PRA0101	Tuesday, 1:00 PM - 2:00 PM	In Person: DH 2020

Refer to ACORN for the most up-to-date information about the location of the course meetings.

Lectures and tutorials start in week 1. See Quercus for details.

Course Contacts

Course Website: <https://q.utoronto.ca/courses/444242>

Teaching Assistant: Seyed Pourya Moghadam Kouhi

Email: pourya.moghadam@mail.utoronto.ca

Office Hours and Location: None

Instructor: Dr. Anthony Bonner

Email: anthony.bonner@utoronto.ca

Phone: 416-997-3463

Office Hours and Location: Tue 3-5pm, DH3084

Additional Notes: Piazza will be used for course communication outside the classroom and office hours. For rare instances, where Piazza cannot be used for communication, emails to the instructor (and TAs) should be sent with "[CSC490] - Topic:" at the beginning of the subject.

Course Overview

This course gives students experience solving a substantial problem that may span several areas of Computer Science. Students will define the scope of the problem, develop a solution plan, produce a working implementation, and present their work using written, oral and (if suitable) video reports. Class time will focus on the project, but may include some lectures. The class will be small and highly interactive. Topics, themes and required preparation will vary by instructor.

This year the course will focus on Machine Learning for Vision. The use of machine learning, and especially deep learning, has led to great progress in solving many difficult problems in machine vision. Example problems include object detection and localization, image segmentation, texture synthesis, scene labelling, colorization, learning artistic style, visual question answering, motion estimation, computational photography, depth estimation, 3D reconstruction and many others.

This capstone design course strives for novel approaches and results in using machine learning to solve problems in machine vision, such as employing a new or advanced combination of a dataset and a machine learning method. Students will work in teams of size 2-4. Teams will be formed in class during the first two to three weeks of the term. Very excellent teams might come up with publishable methods and results.

Course Learning Outcomes

By the end of this course, students will be able to:

1. demonstrate an understanding of the kinds of machine vision problems and data and the variety of available datasets,
2. demonstrate an understanding of the kinds of machine learning methods used in vision and the variety of available software,
3. identify challenging problems in machine learning and vision.
4. download machine learning data and software and apply them to solving machine vision problems,
5. identify a significant problem, understand related approaches, and implement the solution.
6. plan, structure and execute a project productively as a team as well as make significant individual contributions as a team member,
7. communicate ideas, methods, results and give feedback to others.

Prerequisites: Appropriate prerequisite requirement(s) will be available on the UTM timetable along with the topic title prior to course registration.

Corequisites: None

Exclusions: CSC490H1 or CSC491H1

Recommended Preparation: None

Credit Value: 0.5

The course assumes that students have significant programming experience and a solid foundation in the theory and practice of machine learning. Any project in machine learning requires a mastery of training, validation and testing, overfitting and under-fitting, and supervised and unsupervised learning. In addition, state-of-the-art work in machine learning requires neural networks, and machine vision requires convolutional neural networks. Basic methods in machine learning include linear regression and classification, logistic regression, K nearest neighbours, decision trees, probabilistic models, ensemble methods, dimensionality reduction and clustering.

The formal prerequisites are CSC209 and csc311. (Ask for permission from the instructor if any prerequisites are missing.) No special knowledge of machine vision is required.

Course Materials

While searching for and accessing datasets and scientific publications you might experience difficulties because material is behind paywalls or protected by other restrictions. Please make sure, that you are logged in through UofT/UTM library to search for scientific publications. For some datasets, a faculty member has to sign an agreement with the researchers that provide datasets.

Please contact your instructor in those cases.

Marking Scheme

Assessment	Percent	Details	Due Date
In-class Pitch Presentation (individual)	5%	A 3min pitch presentation (slideshow style), where you present two or three machine-vision datasets and associated problems that you would find interesting to work on during this course. Your pitch presentation and the ones from your class mates give you the opportunity to pre-select your group members.	2026-09-15
Report on Problem and Dataset Decision (group)	10%	A written 2-page report is due from each group to clearly state, which data the group will use. Additionally, the report must include details about the machine-learning environment and foreseen methods. The contribution of each team member needs to be stated in the written report. All group members will receive the same grade.	2026-09-29
In-class Progress Presentation (group)	10%	A 10-15min progress presentation as a group (slideshow style). All team members must be active speakers. The contribution of each team member needs to be stated during the presentation. All	2026-11-03

Assessment	Percent	Details	Due Date
		group members will receive the same grade.	
In-class Practice Poster Presentation (individual)	8%	A short presentation where you present your own results in the context of the whole project. You will prepare a poster for this purpose. The presentation will give you the chance to create an initial version of a poster and gather feedback.	2026-11-17
In-class Final Project Poster Presentation (Individual)	4%	In this, second poster presentation, you will include updated results and take advantage of the feedback that you received during the first, practice poster presentation.	2026-12-01
Final Project Implementation and Evaluation (group)	20%	The final project implementation and evaluation assignment comprises a group's work over the whole term. The implementation (source code repository), documentation and evaluation results regarding robustness, quality, novelty, accuracy, etc. will be assessed. The documentation needs to state the individual contributions of the team members. All group members will receive the same grade.	2026-12-08
In-class Brief Reflections (individual)	27%	Almost every week, you have to write a brief reflection (some lines of text or a bullet-point list) on certain aspects of the course, such as your academic integrity, the team formation process and collaboration inside the team, the progress of the project, or peer reviewers' comments on your presentations.	Ongoing

Assessment	Percent	Details	Due Date
In-class Peer Evaluations (individual)	16%	You will give written constructive feedback to some of the other students during their presentations. A grace period of one hour after class/tutorial is allowed for transferring handwritten notes to Quercus.	2026-09-15, 2026-11-03, 2026-11-17, 2026-12-01

Most deadlines for the assignments are due one hour before or after class/tutorial, i.e., Tuesdays at 10am or 3pm. The written feedback or files that are due Tuesdays at 10am will often be needed for in-class discussions or presentations on the same day. You should consider preparing and submitting those assignments and presentation files even earlier (e.g. Mondays).

Your feedback for other students (peer evaluations) and a good portion of the brief reflections are designed in such a way that they can be submitted during class or tutorial. A grace period of one hour is given to transfer hand-written notes to Quercus; thus they are due Tuesdays at 3pm.

Presentations are due at 10am before class as an uploaded file on Quercus, and presented in class the same day. Your participation is needed during the entire lecture and tutorial on these presentation days, because you are asked to give written feedback (peer evaluations, see below) to the other students.

Late Assessment Submissions Policy

No late assignments will be accepted for any reasons.

For all missed term work, either the grades will be reassigned to the final project or additional work will be assigned towards the end of the course, at the discretion of the instructor.

Policies & Statements

Academic Integrity

The Code of Behaviour on Academic Matters states that:

The University and its members have a responsibility to ensure that a climate that might encourage, or conditions that might enable, cheating, misrepresentation, or unfairness is not

tolerated. To this end, all must acknowledge that seeking credit or other advantages by fraud or misrepresentation, or seeking to disadvantage others by disruptive behaviour, is unacceptable, as is any dishonesty or unfairness in dealing with the work or record of a student.

It is your responsibility as a student at the University of Toronto to familiarize yourself with, and adhere to, both the Code of Student Conduct and the Code of Behaviour on Academic Matters. This means, first and foremost, that you should read them carefully.

[Code of Student Conduct](#) and the [Code of Behaviour on Academic Matters](#) are available from the U of T website.

Religious Accommodations

Information about the University's Policy on Scheduling of Classes and Examinations and Other Accommodations for Religious Observances is at

<https://www.vicereprovoctstudents.utoronto.ca/student-resources/rights-responsibilities/accommodation-religious/>

Declaration of Temporary Absence

Students who miss an academic obligation during the term (i.e., in-class assessment, quiz, paper or lab report) may use the ACORN Absence Declaration Tool to record an absence in one or more courses. Students may utilize this option once per term for a single absence period of up to seven consecutive days. The declaration period must include the day of declaration and may include past and/or future dates, for a total of up to 7 calendar days.

Use of the ACORN Absence Declaration does not require supporting documentation and should be used in addition to the missed term work policy outlined in the course syllabus. It remains the student's responsibility to initiate the process for missed academic obligations by following the instructions in the course syllabus.

Re-grading Term Work

A student who believes that their written term work has been unfairly marked may ask the person who marked the work for re-evaluation. Students have up to two weeks from the date of return of an item of term work to inquire about the mark. If the student is not satisfied with this re-evaluation, they may appeal to the instructor in charge of the course if the work was not marked by the instructor (e.g., was marked by a TA). Such re-marking may involve the entire piece of work and may raise or lower the mark. For more information on policies regarding re-marking of term work, please refer to [Re-marking Pieces of Term Work in the Academic Calendar](#).

Use of AI tools

Students may use artificial intelligence tools, including generative AI, in this course as learning aids or to help produce assignments. However, the information provided to the AI tool as prompts must be summarized in an appendix to the work submitted. In addition, any content produced by an AI tool must be cited appropriately. Many organizations that publish standard

citation formats are now providing information on citing generative AI (e.g., MLA: <https://style.mla.org/citing-generative-ai/>).

Students are ultimately accountable for the work they submit, and any mistakes in submitted work, even if produced by an AI tool, may result in a reduced grade.

Additional Content

This course is **writing-intensive**, **speaking-intensive**, and **implementation-intensive** involving several written reports, presentations, reflections, coding sessions, etc. It is recommended that you take advantage of the university's resources, e.g., on effective writing: <https://www.utm.utoronto.ca/asc/writing-resources>

During parts of the lectures and practicals, you will work on and present the progress of your project, listen to other student's presentations, give feedback and get guidance for the next steps with your team members, the instructor and the TAs. It is expected that you attend all lectures and practicals for this purpose. Further individual home and group work is needed to successfully complete your project (e.g. 6 hours additional study time per week).